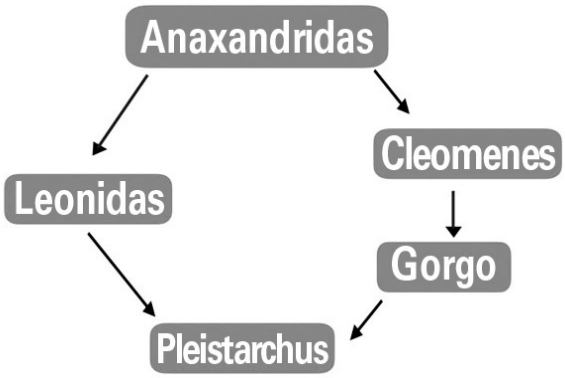
It is like the point where the rainbow touches the forest. We think that we can see it—but if we go to look for it, it isn't there.

If I were to ask, "Are these two stones at the same height?" in interplanetary space, the correct answer would be: "It's a question that doesn't make sense, because there isn't a single notion of 'same height' throughout the universe." If I ask whether two events—one on Earth and the other on Proxima b—are happening "at the same moment," the correct answer would be: "It's a question that doesn't make sense, because there is no such thing as 'the same moment' definable in the universe."

The "present of the universe" is meaningless.

TEMPORAL STRUCTURE WITHOUT THE PRESENT

Gorgo is the woman who saved Greece by realizing that a wax-covered tablet sent there from Persia carried a secret message concealed *beneath* the wax: a message that forewarned the Greeks of a Persian attack. Gorgo had a son called Pleistarchus, fathered by the king of Sparta, the hero of Thermopylae: Leonidas. Leonidas was Gorgo's uncle, the brother of her father, Cleomenes. Who belongs to the "same generation" as Leonidas? Gorgo, who is the mother of his son—or Cleomenes, who is the son of the same father? Here is a diagram for those who, like me, have difficulties with genealogy:



There is an analogy between generations and the temporal structure of the world as revealed by relativity. It makes no sense to ask if it is Cleomenes or Gorgo who is "of the same generation" as Leonidas, because there is no single concept[30] of "same generation." If we say that Leonidas and his brother are "of the same generation" because they have the same father, and that Leonidas and his wife are "of the same generation" because they have a son together, we must therefore say that this "same generation" includes Gorgo and her own father! The filial relationship establishes an order between